



COMPETENCY STANDARD

FOR

ELECTRICAL INSTALLATION AND MAINTENANCE

(SHIPBUILDING SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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The Competency Standards for Electrical Installation and Maintenance (Shipbuilding) serve as a foundational document for the development of curricula, teaching materials, and assessment tools specifically tailored for the Shipbuilding sector. This document ensures that training activities align with industry requirements, enabling individuals who successfully meet the established standards through assessment to become qualified professionals in relevant roles within the sector.

Developed under the Skills for Industry Competitiveness and Innovation Program (SICIP), this document addresses the skills requirements critical for Electrical Installation and Maintenance. It is owned by the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

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Skills for Industry Competitiveness and Innovation Program (SICIP)

Finance Division, Ministry of Finance,

Probashi Kallyan Bhaban (Level-15 & 16)

71-72 Old Elephant Road, Eskaton Garden

Dhaka-1000

Phone: +8802 55138753-55, Fax: +88 02 55138752

Website: <https://sicip.gov.bd/>

INTRODUCTION:

The **Skills for Industry Competitiveness and Innovation Program (SICIP)** has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution and relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Identification and validation of units of competency and elements for each occupation were made by experts from Construction industry in consultative workshop.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.

- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:

Generic Competencies (60 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-SBD-EIM-01-G	Apply occupational health and safety (OHS) practice in the workplace	1. Identify OHS policies and procedures. 2. Apply personal health and safety practices. 3. Report hazards and risks 4. Respond to emergencies.	20
SICIP-SBD-EIM-02-G	Perform Computations Using Basic Mathematical Concepts	1. Identify calculation requirements in the workplace 2. Select appropriate mathematical methods 3. Perform calculations	20
SICIP-SBD-EIM-03-G	Interpret Technical Drawings and Manuals	1. Select technical drawing 2. Interpret technical drawings. 3. Interpret operation & maintenance manuals	20
Total Hour			60 Hrs.

Sector Specific Competencies (50 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-SBD-EIM-01-S	Apply Green Practices in Shipbuilding sector	1. Interpret green concepts 2. Minimize resource use in the workplace 3. Implement waste management practices	10
SICIP-SBD-EIM-02-S	Use Hand and Power Tools	1. Identify and inspect hand and power tools 2. Use hand tools properly and safely 3. Operate power tools properly and safely 4. Clean and maintain hand tools and power tools	20
SICIP-SBD-EIM-03-S	Carry Out Precision Checks and Measurements	1. Select the job to be checked and measured 2. Select measuring and checking tool/instrument 3. Perform measurements 4. Check and record measurement 5. Clean, maintain and store measuring instruments.	20
Total Hour			50 Hrs.

Occupation Specific Competencies (370 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-SBD-EIM-01-O	Understand Basic Electrical System	1 Interpret different types of cables used in ship. 2 Understand different types of electrical loads and load calculation 3 Develop basic electrical circuit 4 Perform channel wiring 5 Perform Cable Laying and Securing in a ship	80
SICIP-SBD-EIM-02-O	Perform Motor Connection	1 Identify and select controlling and protective devices for motor connection 2 Install controlling and protective devices 3 Install and operate single phase Induction motor 4 Install and operate three phase Induction motor	90
SICIP-SBD-EIM-03-O	Operate Three Phase Generator	1 Install, connect and repair Automatic Transfer Switch (ATS) 2 Connect the generator to the main electrical panel 3 Prepare generator for operation 4 Start and stop generator 5 Perform post-operation checks and maintenance	70
SICIP-SBD-EIM-04-O	Perform Single Phase Motor Rewinding and Servicing	1 Check single phase motor and detect fault 2 Select tools and prepare material for winding / rewinding 3 Carryout winding / rewinding of stator rotor and armature 4 Make connections, carry out pre-assembly tests	60

Code	Unit of Competency	Elements of Competency	Duration (hours)
		and assemble the machine 5 Carry out final test 6 Record the test result	
SICIP-SBD-EIM-05-O	Perform Three Phase Motor Rewinding and Servicing	1 Check three phase motor and detect fault 2 Select tools and prepare material for winding / rewinding 3 Carryout winding / rewinding of stator rotor 4 Make connections, carry out pre-assembly tests and assemble the machine 5 Carry out final test 6 Record the test result	70
Total Hours			0 Hrs.

COMPETENCY STANDARD: ELECTRICAL INSTALLATION AND MAINTENANCE (SHIPBUILDING)

Generic Competencies

Unit of Competency: APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICE IN THE WORKPLACE	Nominal Duration: 20 hrs.	Unit Code: SICIP-SBD-EIM-01-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply occupational health and safety (OHS) practices in the workplace. It specifically includes the tasks of identifying OHS policies and procedures, applying personal health and safety practices, reporting hazards and risks and responding to emergencies.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1.1 <u>OHS policies</u> and safe operating procedures are interpreted 1.2 Safety signs and symbols are identified and followed. 1.3 Response, evacuation procedures and other contingency measures are interpreted correctly.
2. Apply personal health and safety practices	1.4 OHS policies and procedures are interpreted in the workplace including <u>personal protective equipment (PPE)</u> . 1.5 Common health issues are recognized. 1.6 Common safety issues are identified and applied.
3. Report hazards and risks	1.1 Hazards and risks are identified. 1.2 Hazards and risks assessment and controls are interpreted 1.3 Hazards and risk assessment are reported
4. Respond to emergencies	1.1 Respond to alarms and warning devices. 1.2 <u>Emergency response plans and procedures</u> are responded to. 1.3 <u>First aid procedures</u> during emergency situations are identified.

Range of variables:

Variables	Range (may include but not limited to)
1. OHS policies	1.1 Organizational OHS policies 1.2 International OHS requirements 1.3 Fire safety rules and regulations
2. Personal protective	2.1 Safety glasses

equipment	2.2 Ear plugs 2.3 Gloves 2.4 Apron 2.5 Helmet 2.6 Mask 2.7 Safety shoes
3. Emergency response plans and procedures	3.1 Firefighting procedures 3.2 Earthquake response procedures 3.3 Emergency response plans and procedures 3.4 Medical and first aid
4. First aid procedure	4.1 Washing of open wound 4.2 Washing chemically infected area 4.3 Applying bandage 4.4 Taking appropriate medicine

Curricular Content Guide

1. Underpinning knowledge	1.1 Workplace OHS policies and procedures 1.2 Work safety procedures 1.3 Emergency response procedures: 1.3.1 Firefighting 1.3.2 Earthquake response 1.3.3 Accident response 1.4 Types of hazards (biological, chemical and physical) and their effects 1.5 OHS awareness 1.6 Personal protective equipment (PPE) 1.7 Malfunctions and resolutions
2. Underpinning skill	1.8 Identifying OHS policies and procedures 1.9 Applying personal health and safety practices 1.10 Reporting hazards and risks 1.11 Responding to emergencies
3. Underpinning Attitudes	1.12 Commitment to occupational health and safety 1.13 Promptness in carrying out activities 1.14 Sincere and honest to duties 1.15 Environmental concerns 1.16 Eagerness to learn 1.17 Tidiness and timeliness 1.18 Respect for rights of peers and seniors in the workplace 1.19 Communication with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 1.20 Workplace (simulated or actual) 1.21 Workplace documents, signs and symbols 1.22 Codes of conduct 1.23 Projector 1.24 Learning manual

	1.25 Tools, equipment and facilities appropriate to the process or activities. 1.26 Materials are relevant to the proposed activity.
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Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 identified OHS policies and procedures 1.2 applied personal health and safety practices (including PPE) 1.3 reported hazards and risks 1.4 responded to emergencies.
2. Methods of assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM COMPUTATIONS USING BASIC MATHEMATICAL CONCEPTS	Nominal Duration: 20 hrs.	Unit Code: SICIP-SBD-EIM-02-G
Unit Descriptor: This unit of competency requires knowledge, skills and attitude to perform computations using basic mathematical concepts in workplace. It specifically includes the tasks of identifying calculation requirements in the workplace, selecting appropriate mathematical methods, and performing calculations.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify calculation requirements in the workplace	1.1 <u>Calculation requirements</u> are identified based on the tasks and specifications. 1.2 Calculation requirements are understood. 1.3 <u>Workplace information</u> is interpreted to determine the necessary calculations. 1.4 Relevant workplace documents are identified to gather necessary data for calculations.
2. Select appropriate mathematical methods	2.1 <u>Appropriate calculation method</u> is identified based on the specific requirements of the task. 2.2 Units of measurement are considered when selecting the method for conversion. 2.3 Selected calculation method is applied to verify its effectiveness before proceeding with the final calculation.
3. Perform calculations	3.1 Appropriate <u>tools and instruments</u> are identified based on the requirements of the calculation. 3.2 Tools and instruments are checked for accuracy and calibration. 3.3 Computation is made using the identified tools and instruments. 3.4 Final result is checked to verify the correctness of the calculation before reporting.

Range of variables:

Variables	Range (may include but not limited to)
1. Calculation requirements.	1.1 Area 1.2 Volume 1.3 Measurements: 1.3.1 Length 1.3.2 Width

	1.3.3 Height 1.3.4 Thickness 1.3.5 Diameter
2. Workplace information	2.1 Drawing 2.2 Part drawing 2.3 Verbal instructions 2.4 Job order and Job sheet
3. Appropriate calculation method	3.1 Addition 3.2 Subtraction 3.3 Division and Multiplication 3.4 Conversion of measurement 3.5 Percentage and ratio calculation 3.6 Geometric Formulas (Square, Rectangle, Circle, Sphere, etc.)
4. Tools and instruments	4.1 Calculator 4.2 Computer

Curricular Content Guide

1. Underpinning knowledge	1.1 Numerical concept, identification of calculation requirements 1.2 Basic mathematical methods, such as addition, subtraction, multiplication, division, percentage, conversion, ratio and basic geometry formulas 1.3 Mathematical language, symbols 1.4 Measuring units
2. Underpinning skill	2.1 Computing basic mathematics 2.2 Practicing conversion of measurements 2.3 Calculating areas, volume, etc. using formulas
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Codes of conduct 4.5 Projector 4.6 Learning manual 4.7 Tools, equipment and facilities appropriate to the process

	or activities. 4.8 Materials are relevant to the proposed activity.
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Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 identified calculation requirements from workplace information. 1.2 selected appropriate method to carry out the calculation requirements. 1.3 completed calculations using appropriate tools/instruments.
2. Methods of assessment	Methods of assessment may include but not limited to: 1.4 Written test 1.5 Practical Demonstration 1.6 Oral Questioning 1.7 Portfolio (Optional)
3. Context of assessment	1.8 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 1.9 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: INTERPRET TECHNICAL DRAWINGS AND MANUALS	Nominal Duration: 20 hrs.	Unit Code: SICIP-SBD-EIM-03-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to translate technical drawings and plans. It specifically includes the tasks of selecting technical drawing, interpreting operation and maintenance manuals.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Select technical drawing	1.1 Appropriate <u>technical drawing</u> is selected based on the specifications and requirements of the job. 1.2 Selected drawing is checked to ensure that it accurately reflects the job requirements.
2. Interpret technical drawings.	2.1 Drawing components, assemblies are identified 2.2 Dimensions are identified according to job requirement 2.3 Clearances/tolerances are checked in accordance with workplace standard 2.4 <u>Instructions</u> are identified and followed accurately. 2.5 Material <u>specifications</u> are interpreted 2.6 Symbols in drawing are interpreted.
3. Interpret operation & maintenance manuals	3.1 Operation and maintenance manuals are collected and interpreted 3.2 Operation and maintenance manuals are followed.

Range of variables:

Variables	Range (may include but not limited to)
1. Technical drawing	1.1 Sketches 1.2 Manuals
2. Instructions	2.1 Note 2.2 Instruction 2.3 Special instruction 2.4 Precaution
3. Specifications	3.1 Product specifications 3.2 Method specifications 3.3 Material specifications

Curricular Content Guide

1. Underpinning knowledge	1.1 Technical drawing interpretation 1.2 Sequence of drawing 1.3 Methods of checking and applying drawing for work 1.4 Drawing selection and checking method to ensure
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	conformity to the job requirements. 1.5 Drawing components, assemblies 1.6 Identification of dimensions according to job requirement 1.7 Procedure of checking clearances/tolerances 1.8 Work instructions 1.9 Material specifications 1.10 Drawing symbols interpretation 1.11 Use of operation and maintenance manuals
2. Underpinning skill	2.1 Practicing workplace safety 2.2 Interpreting drawing, following operation and maintenance manuals, 2.3 Performing jobs in accordance with the drawing 2.4 Performing calculation as per drawing 2.5 Selecting and checking of drawing to ensure conformity to the job requirements. 2.6 Identifying drawing components and assemblies 2.7 Identifying dimensions according to job requirement 2.8 Checking clearances/tolerances in accordance with workplace standard 2.9 Following operation and maintenance manuals
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Eagerness to learn 3.5 Tidiness and timeliness 3.6 Environmental concerns 3.7 Respect for rights of peers and seniors at workplace 3.8 Communication with peers and seniors at workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Workplace documents, Projector 4.3 Learning manual 4.4 Tools, equipment and facilities appropriate to the process or activities. 4.5 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 identified dimension according to job requirement 1.2 maintained clearances and tolerances according to workplace requirement. 1.3 interpreted technical drawing 1.4 interpreted operation & maintenance manuals
2. Methods of assessment	Methods of assessment may include but not limited to: 2.1 Written test

	2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Sector Specific Competencies

Unit of Competency: APPLY GREEN PRACTICES IN SHIPBUILDING SECTOR	Nominal Duration: 10 hrs.	Unit Code: SICIP-SBD-EIM-01-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply green practices in shipbuilding sector. It specifically includes the tasks of interpreting green concepts, minimizing resources use in the workplace and implementing waste management practices.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1.1 <u>Basic concepts of green</u> are understood. 1.2 Energy efficiency principles are interpreted. 1.3 <u>Sustainable materials</u> are identified. 1.4 Pollution control methods are understood. 1.5 <u>Environmental certifications and standards</u> are identified. 1.6 <u>Green supply chains</u> are interpreted to minimize environmental impact.
2. Minimize resources use in the workplace	2.1 Resource consumption is identified in the Shipbuilding sector. 2.2 <u>Energy-efficient equipment</u> is identified and maintained. 2.3 Materials are sourced ensuring that sustainable and recyclable materials are prioritized. 2.4 Water usage is minimized through efficient management. 2.5 Production processes are optimized. 2.6 <u>Green technologies and innovations</u> are implemented in the workplace.
3. Implement waste management practices	3.1 <u>Waste types</u> identified in the Shipbuilding sector. 3.2 Waste segregation practices are interpreted. 3.3 Recycling programs are established through external recycling facilities. 3.4 <u>Waste reduction strategies</u> are understood. 3.5 Safe disposal practices are described. 3.6 <u>Waste-to-energy solutions</u> are explored to reduce landfill contributions.

Range of Variables

Variable	Range May include but not limited to:
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1. Basic concepts of green	1.1 Principles of Green Practices 1.2 Sources of Environmental Impacts 1.3 Resource Utilization 1.4 Sustainable Alternatives 1.5 Waste Management Practices 1.6 Green Habits at Work 1.7 Compliance and Regulations
2. Sustainable materials	2.1 Recycled Metals 2.2 Biodegradable Lubricants and Fluids 2.3 Composite Materials with Low Environmental Impact 2.4 Low-VOC Paints and Coatings 2.5 Eco-friendly Packaging Materials 2.6 Durable and Repairable Components
3. Environmental certifications and standards	3.1 ISO 14001: Environmental Management System (EMS) 3.2 LEED Certification (Leadership in Energy and Environmental Design) 3.3 RoHS (Restriction of Hazardous Substances) 3.4 REACH (Registration, Evaluation, Authorization and Restriction of Chemicals) 3.5 Energy Star Certification 3.6 Eco-Labeling Standards 3.7 Green Building Codes & Local Environmental Compliance
4. Green supply chains	4.1 Environmentally Responsible Sourcing 4.2 Eco-Friendly Transportation and Logistics 4.3 Supplier Environmental Compliance 4.4 Sustainable Packaging and Distribution
5. Energy-efficient equipment	5.1 High-Efficiency Electric Motors (IE2/IE3/IE4 Rated) 5.2 Variable Frequency Drives (VFDs) 5.3 Energy-Efficient LED Lighting Systems 5.4 Energy-Star Rated Industrial Equipment 5.5 Heat Recovery and Insulation Systems
6. Green technologies and innovations	6.1 Renewable Energy Integration 6.2 Additive Manufacturing (3D Printing) 6.3 Smart Manufacturing Systems (Industry 4.0) 6.4 Waste-to-Energy Technologies 6.5 Low-Emission Welding and Cutting Equipment 6.6 Closed-Loop Cooling and Lubrication Systems 6.7 Eco-Innovation in Product Design
7. Waste types	7.1 Metal Waste 7.2 Plastic and Packaging Waste 7.3 Hazardous Waste 7.4 Electronic Waste (E-Waste) 7.5 Dust and Particulate Waste 7.6 Wood and Pallet Waste
8. Waste reduction strategies	8.1 5S Workplace Organization 8.2 Lean Manufacturing Practices

	8.3 Material Optimization and Nesting Techniques 8.4 Reusing and Repurposing Waste
9. Waste-to-energy solutions	9.1 Thermal Conversion (Incineration with Energy Recovery) 9.2 Gasification and Pyrolysis 9.3 Biomass Boilers Using Organic Waste 9.4 Use of Waste Oils as Alternative Fuel 9.5 Waste Heat Recovery Systems (WHRS) 9.6 On-Site Micro Energy Generation Units

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of pollution. 1.2 Pollution control technologies 1.3 Legal and environmental frameworks: overview of national environmental policies and local environmental compliance regulations. 1.4 Importance of ISO 14001 and other EMS (Environmental Management System) standards. 1.5 Classification of sustainable, recyclable, and hazardous materials used in production. 1.6 Strategies for minimizing water and energy use during material processing. 1.7 Waste hierarchy: reduce, reuse, recycle, recover, and disposal (5Rs). 1.8 Industrial waste-to-energy concepts.
2. Underpinning Skills	2.1 Measuring and analyzing energy and water consumption using meters. 2.2 Identifying and operating energy-efficient machinery. 2.3 Monitoring and logging resource consumption data. 2.4 Adjusting machine settings or workflow layouts. 2.5 Applying green inventory practices. 2.6 Segregating waste effectively into recyclable, hazardous, and general categories. 2.7 Implementing basic pollution control activities.
3 Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Stationery

	4.4 Learning manual
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Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted green concepts 1.2 minimized resources use in the workplace 1.3 implementing waste management practices
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: USE HAND AND POWER TOOLS	Nominal Duration: 20 hrs.	Unit Code: SICIP-SBD-EIM-02-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to use hand and power tools in the workplace. It specifically includes the task of identifying and inspecting hand and power tools, using hand tools properly and safely, operating power tools properly and safely and cleaning and maintaining hand and power tools.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify and inspect hand and power tools	1.1 Appropriate hand and power tools are identified. 1.2 Application of hand and power tools is recognized. 1.3 Usability of hand and power tools is checked and verified.
2. Use hand tools properly and safely	2.1 Appropriate <u>hand tools</u> are selected. 2.2 Safety precautions are ensured before using hand tools. 2.3 Unsafe or faulty hand tools are identified and marked for repair. 2.4 <u>Measuring tools</u> are checked and calibrated before use. 2.5 Use hand tools properly and safely to perform work activity.
3. Operate power tools properly and safely	3.1. Appropriate <u>power tools</u> are selected. 3.2. Power supply outlet and electrical cord are inspected and confirmed safe for use in accordance with established workplace safety requirements. 3.3. Safety precautions are ensured before using power tools in accordance with manufacturer's operating specification. 3.4. Proper sequence of operation is maintained using power tools. 3.5. Unsafe or faulty power tools are identified and marked for repair. 3.6. Power tools are operated properly and safely to perform bench work.
4. Clean and maintain hand and power tools	4.1. Dust and foreign matter are removed from hand and power tools in accordance to workplace standards. 4.2. Condition of hand and power tools is checked after use and reported. 4.3. Appropriate lubricant is applied after use and prior to storage.

	4.4. Defective hand and power tools are inspected and repaired or replaced. 4.5. Hand and power tools are stored and secured in accordance with workplace requirements.
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Range of Variables

Variable	Range May include but not limited to:
1. Hand tools	1.1 Hammer 1.2 Bench vice 1.3 Files 1.4 Wrenches 1.5 Pliers 1.6 Scriber 1.7 Scraper Screw 1.8 Surface plate 1.9 Gauge 1.10 Tap sets 1.11 Die sets 1.12 Hacksaw 1.13 Spanners 1.14 Vice grip 1.15 Wire cutters 1.16 Clamps 1.17 Jacks
2. Power tools	2.1 Drills 2.2 Rivet gun 2.3 Grinders 2.4 Pneumatic wrenches 2.5 Press machine 2.6 Cutting 2.7 Saws 2.8 Soldering iron
3. Measuring tools	3.1 Micrometer 3.2 Megger 3.3 Measuring tape 3.4 Hose level 3.5 Water level 3.6 Calipers 3.7 Steel rule 3.8 Meter rule 3.9 Spirit level 3.10 Protractor 3.11 Tri-square 3.12 Gauges

Curricular Content Guide

1. Underpinning Knowledge	1.1 Information on hand and power tools, their functions and use 1.2 Procedures for safely using hand and power tools
2. Underpinning Skills	2.1 Identifying hand, power and measuring tools 2.2 Following safety precautions when using hand, power and measuring tools 2.3 Using hand and measuring tools correctly and safely in accordance with manufacturer's operating specification 2.4 Operating power tools correctly and safely in accordance with manufacturer's operating specification 2.5 Cleaning and maintaining hand and power tools after use 2.6 Applying appropriate lubricant on hand and power tools after use and prior to storing
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Codes of conduct 4.5 Projector 4.6 Learning manual 4.7 Tools, equipment and facilities appropriate to the process or activities. 4.8 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified and selected appropriate hand and power tools for work to be performed 1.2 identified and used measuring and testing tools appropriate to work activity 1.3 followed safety precautions when using hand and power tools 1.4 operated power tools safely and pursuant to
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	manufacturer's operating specification 1.5 performed cleaning and maintenance of hand and power tools after use and prior to storing
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: CARRY OUT PRECISION CHECKS AND MEASUREMENTS	Nominal Duration: 20 hrs.	Unit Code: SICIP-SBD-EIM-03-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to carry out precision checks and measurements. It specifically includes the tasks of selecting the job to be measured, selecting measuring and checking tools/instrument, performing measurements, checking & recording measurements and cleaning, maintaining and storing measuring instruments.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Select the job to be checked and measured	1.1 Job is selected for measuring and checking 1.2 Required <u>dimensional measurement</u> is determined in accordance with drawing/plan 1.3 Required <u>physical conditions</u> are identified in accordance with drawing/plan 1.4 Required <u>geometrical dimensions</u> are identified in accordance with drawing/plan 1.5 Job drawing is used to select the measuring instruments.
2. Select measuring and checking tool/instrument	2.1 Appropriate measuring instruments is selected in accordance with job requirement. 2.2 <u>Direct and indirect measuring instruments</u> and <u>checking instrument</u> are identified 2.3 Applications of measuring device is determined. 2.4 Usability and accuracy of measuring device is checked and verified. 2.5 Measuring device is prepared for measurement. 2.6 Fits, Tolerance, clearance and limits are identified according to job requirements.
3. Perform measurements	3.1 Measurements are obtained using appropriate measuring instrument. 3.2 <u>Systems of measurements</u> are identified and converted where necessary. 3.3 Measurement is kept accurately in accordance to specification 3.4 Measurement is checked against job requirement 3.5 Physical conditions are checked in accordance with job requirements 3.6 Geometrical dimensions are checked in accordance with job specifications
4. Check and record	4.1 Measurements are checked and recorded in

measurement	accordance with workplace procedure 4.2 Measurement is interpreted, recorded and communicated to authority
5. Clean, maintain and store measuring instruments.	5.1 Dust and dirt are removed from the measuring instruments 5.2 Condition of measuring instruments are checked 5.3 Appropriate lubricant is applied after use and prior to storage 5.4 Measuring instruments are checked and calibrated 5.5 Measuring instruments are stored in accordance with workplace procedure.

Range of Variables

Variable	Range May include but not limited to:
1. Dimensional measurement	1.1 Length 1.2 Width 1.3 Depth 1.4 Diameter 1.5 Radius 1.6 Height
2. Physical condition	2.1 Roughness 2.2 Color 2.3 Smoothness 2.4 Surface finish 2.5 Flatness
3. Geometrical dimension	3.1 Angle 3.2 Circle 3.3 Triangle 3.4 Square 3.5 Rectangle 3.6 Hexagon shape 3.7 Pentagon shape
4. Direct measuring instruments	4.1 Set squares 4.2 Dial indicators 4.3 Steel tape 4.4 Steel rule 4.5 Meter rule 4.6 Calculator 4.7 Vernier slide caliper 4.8 Digital Vernier slide caliper 4.9 Micrometer (inch/millimeter) 4.10 Digital micrometer 4.11 Vernier bevel protractor 4.12 Spirit level 4.13 AVO meter(analogue/digital)

	4.14 Thermometers 4.15 Water meter 4.16 Gas meter 4.17 Simple protractor
5. Indirect measuring instrument	5.1 Outside caliper 5.2 Inside caliper 5.3 Bevel tri-square 5.4 Telescoping gage 5.5 Straight edge 5.6 Side bar 5.7 Trammel
6. Checking instrument.	6.1 Plug gauge 6.2 Snap gauge 6.3 Screw pitch gauge 6.4 Slip gauges 6.5 Feeler gauges 6.6 Screw pitch gauge 6.7 Slip gauge 6.8 Tri-square 6.9 Center gauge 6.10 Bevel tri-square
7. Systems of measurements	7.1 ISO standard 7.2 English system 7.3 Metric system

Curricular Content Guide

1. Underpinning Knowledge	1.1 Difference between measuring and checking 1.2 Types of measuring tools and their applications 1.3 Types of checking tools and their applications 1.4 Geometrical dimensions and tolerances 1.5 Method, procedure and techniques when taking linear Measurements 1.6 Methods, procedures and techniques when checking physical conditions of work pieces 1.7 Methods, procedures and techniques when checking geometrical dimensions of work pieces 1.8 Measurement conversion systems 1.9 Workplace record keeping procedures 1.10 Preventive maintenance for measuring and checking tools 1.11 Calibration and adjustment procedures for measuring and checking tools
2. Underpinning Skills	2.1 Determining required dimensional measurements, physical conditions and geometrical dimensions in accordance with drawing/plan and workplace specification 2.2 Measuring and checking linear and geometrical

	dimensions within the required tolerance in accordance to specification 2.3 Checking physical conditions using appropriate checking tool 2.4 Identifying and converting systems of measurements where necessary. 2.5 Recording measurements in accordance with workplace procedure 2.6 Interpreting and communicating measurement to authority 2.7 Applying appropriate lubricant on measuring and checking tools and instruments after use and prior to storage 2.8 Checking condition of measuring instruments, calibrating and storing in accordance with workplace procedure
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Codes of conduct 4.5 Projector 4.6 Learning manual 4.7 Tools, equipment and facilities appropriate to the process or activities. 4.8 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 determined required dimensional measurements, physical conditions and geometrical dimensions in accordance with drawing/plan and workplace specification 1.2 measured and checked linear and geometrical dimensions within the required tolerance in accordance to specification 1.3 checked physical conditions using appropriate checking
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	tool 1.4 identified and converted systems of measurements where necessary. 1.5 recorded measurements in accordance with workplace procedure 1.6 interpreted and communicated measurement to authority 1.7 applied appropriate lubricant on measuring and checking tools and instruments after use and prior to storage 1.8 checked condition of measuring instruments, calibrated and stored in accordance with workplace procedure
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Occupation Specific Competencies

Unit of Competency: UNDERSTAND BASIC ELECTRICAL SYSTEM	Nominal Duration: 80 Hrs.	Unit Code: SICIP-SBD-EIM-01-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to understand basic electrical system. It specifically includes the task of interpreting different types of cables used in ship, understanding different types of electrical loads and load calculation, developing basic electrical circuit, performing channel wiring, performing cable Laying and securing in a ship.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret different types of cables used in ship	1.1. <u>Different types of cables</u> used in ship are explained 1.2. <u>Characteristics and size</u> of the cables are interpreted.
2. Understand different types of electrical loads and load calculation	2.1. <u>Resistive loads</u> are identified and explained 2.2. <u>Inductive loads</u> are identified and explained 2.3. <u>Capacitive loads</u> are identified and explained 2.4. <u>Ship accommodation, ship bridge, ship galley and Steering gear rooms load</u> is calculated
3. Develop basic electrical circuit	3.1. The <u>basic Electrical Parameters with Units</u> of electricity are understood and interpreted. 3.2. Clear and accurate circuit diagrams for series, parallel, and series-parallel circuits are developed using source, load, controlling, and protective devices. 3.3. Series circuits are assembled following the designed circuit diagram. 3.4. Parallel circuits are assembled following the designed circuit diagram. 3.5. Series-parallel circuits are assembled following the designed circuit diagram. 3.6. Circuit parameters are measured and analyzed. 3.7. Troubleshooting and necessary adjustments are made to enhance functionality or reliability as required.
4. Perform channel wiring	4.1. The wiring board, tools, and materials are prepared according to the job specifications and wiring diagram. 4.2. Channels are securely fixed on the board as per the layout, ensuring proper alignment. 4.3. Cables are drawn through channels, avoiding damage and ensuring correct routing as specified. 4.4. Electrical components, such as switches, sockets, regulators, and ballast, are installed and securely fitted

	on the board. 4.5 All components and circuits are connected accurately following the wiring diagram and specifications. 4.6 Protective devices, including circuit breakers and fuses, are installed and connected to ensure safety. 4.7 The upper covers are securely placed and attached on the channels after wiring is completed. 4.8 The completed wiring is tested for functionality.
5. Perform Cable Laying and Securing in a ship	5.1 The layout of cable trays is completed as per approved drawings and specifications. 5.2 Cable trays are installed securely and in compliance with safety and regulatory standards. 5.3 Cables are laid along specified routes, ensuring proper handling to avoid damage. 5.4 Main Distribution Board (MDB) and Sub Distribution Board (SDB) are assembled with Circuit Breakers, Bus-bar, and indication lamps. 5.5 Cables are connected with the Main Distribution Board (MDB) and Sub Distribution Board (SDB). 5.6 Communication and power cables are separated and installed to minimize electromagnetic interference. 5.7 Appropriate protection devices, such as cable glands and fire barriers, are installed to safeguard the system. 5.8 Equipment and loads are connected as per wiring diagrams and tested for functionality.

Range of Variables

Variable	Range (Includes but not limited to):
1. Different types of cables	1.1 Flat/Round marine cable 1.2 Coaxial cable 1.3 Fire resistant cable 1.4 Marine power cable 1.5 Control Cable 1.6 Marine battery cable 1.7 Marine wire 1.8 Ship telephone cable
2. Characteristics and size of the cables	Characteristics 2.1 Conductivity 2.2 Insulation Resistance 2.3 Current and Voltage Capacity 2.4 Single/Multicore 2.5 Color Code 2.6 Skin Effect Size

	2.7 1.5rm, 2.5rm, 4rm, 6rm, 10rm, 16rm, 25rm, 35rm, 50rm, 95rm, 120rm, 135rm.
3. Resistive loads	3.1 Electric bulb 3.2 Heater coil 3.3 Electric Iron 3.4 Rice Cooker
4. Inductive loads	4.1 Induction motor 4.2 Electric fan 4.3 Magnetic Conductor 4.4 Electric choke coil/Ballast
5. Capacitive loads	5.1 Capacitor load bank 5.2 PFI plant
6. Ship accommodation load	6.1 Light 6.2 AC 6.3 Fan 6.4 Computer 6.5 Printer
7. Ship bridge load	7.1 Gyro compass 7.2 General broadcast system 7.3 Fire alarm 7.4 VHF 7.5 GPS 7.6 Echo sounder
8. Ship galley load	8.1 Hot Plate 8.2 Rice Cooker 8.3 Blender 8.4 Exhaust Blower 8.5 Micro Oven 8.6 Refrigerator 8.7 Heater
9. Basic Electrical Parameters with Units	9.1 Voltage, Unit-Volt 9.2 Current, Unit- Ampere 9.3 Resistance, unit-Ohm 9.4 Power, Unit-Watt
10. types of protection devices	10.1 Fuse 10.2 MCB 10.3 MCCB 10.4 ACB 10.5 Lighting arrestor
11. Cable Tray	11.1 Steel cable tray 11.2 Aluminum cable tray 11.3 PVC cable tray
12. Equipment and different loads	12.1 Generator 12.2 Transformer 12.3 Chiller 12.4 Boiler 12.5 AC/DC Motor

	12.6 RADAR 12.7 Echo Sounder 12.8 GPS
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Curricular Content Guide

1. Underpinning Knowledge	1.1. Types, characteristics, and uses of cables in ships. 1.2. Fundamentals of resistive, inductive, and capacitive loads. 1.3. Principles of series, parallel, and series-parallel circuits. 1.4. Reading and interpreting wiring diagrams. 1.5. Parameters affecting circuit performance (resistance, voltage, current). 1.6. Safety standards and regulations for marine electrical systems. 1.7. Proper handling and protection of electrical equipment (e.g., fire barriers, circuit breakers). 1.8. Procedures for fitting electrical components (boards, sockets, switches). 1.1 Standards for cable tray layouts, securing, and minimizing electromagnetic interference.
2. Underpinning Skills	2.1 Interpret technical documents, including electrical diagrams and layouts. 2.2 Design and assemble series, parallel, and series-parallel circuits. 2.3 Measure, analyze, and troubleshoot circuit parameters. 2.4 Install and connect electrical boards, switches, sockets, and accessories. 2.5 Perform cable laying and securing on ships. 2.6 Conduct functionality testing and fault resolution for electrical systems.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety. 3.2 Promptness in carrying out activities. 3.3 Sincere & honest to duties. 3.4 Eagerness to learn. 3.5 Tidiness & timeliness. 3.6 Environmental concerns. 3.7 Respect for rights of peers and seniors at workplace. 3.8 Communication with peers and seniors at workplace
4 Resource Implications	The following resources must be provided: 4.1 Workplace (actual or simulated) 4.2 Tools and equipment appropriate to joint 4.3 Accessibility to the work place 4.4 Quality materials availability CBLM

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted different types of cables used in ship 1.2 understood different types of electrical loads and load calculation 1.3 developed basic electrical circuit 1.4 performed channel wiring 1.5 performed cable laying and securing in a ship
2. Methods of Assessment	2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM MOTOR CONNECTION	Nominal Duration: 90 Hrs.	Unit Code: SICIP-SBD-EIM-02-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform motor connection. It specially includes the task of identifying and selecting controlling and protective devices for motor connection, installing controlling and protective devices, installing and operating single phase Induction motor, installing and operating three phase induction motor.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify and select controlling and protective devices for motor connection	1.1 <u>Manuals</u> and documents of controlling and protective devices are collected. 1.2 <u>Electric Diagram</u> of controlling and protective devices are sorted. 1.3 Types of <u>controlling</u> and <u>protective</u> devices are listed
2. Install controlling and protective devices	2.1 Controlling and protective devices are selected and collected according to the need of the operations 2.2 Controlling and protective devices are installed according to the layout plan 2.3 Controlling and protective devices are set and connected to the motor
3. Install and operate single phase Induction motor	3.1. The <u>motor's</u> rated voltage, current, and power specifications are checked before installation. 3.2. Appropriate type and size of cable are identified and selected according to load and circuit requirements. 3.3. Insulation is stripped, and cable lugs are crimped securely for motor and supply connections. 3.4. The live/phase, neutral, and earth wires are connected to the motor's terminal block according to the wiring diagram. 3.5. Motor control switch is connected as per safety standards. 3.6. All connections are tightened, and the motor is tested and operated. 3.7. The motor's rotation direction is verified and adjusted if necessary to ensure correct functionality.
4. Install and operate three phase Induction motor	4.1 The motor's rated voltage, current, and power specifications are verified before installation. 4.2 Appropriate type and size of cable and Circuit Breaker are calculated, identified and selected based on motor load 4.3 Insulation is stripped, and cable lugs are crimped

	securely for the motor and supply connections. 4.4 For star connection, the three phase wires are connected to the motor terminals as per the star configuration diagram. 4.5 For delta connection, the three phase wires are connected according to the delta configuration diagram. 4.6 <u>Starter</u> are connected to the motor terminals, including overload protection and control circuits. 4.7 The motor is tested, and rotation direction is verified and adjusted if necessary to ensure correct functionality.
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Range of Variables

Variable	Range (Includes but not limited to):
1. Manuals	1.1 Manufacturer's specification manual 1.2 Troubleshoot manual 1.3 Maintenance manual
2. Electric Diagrams	2.1 Wiring drawings 2.2 Control diagram
3. Controlling Devices	3.1 Manual Switch 3.2 Push-Button Switch 3.3 Direct On Line (DOL) Starter 3.4 Reversing Starter 3.5 Timers 3.6 Control Relays 3.7 Overload Relays 3.8 Variable Frequency Drives (VFDs)
4. Protective devices	4.1 Thermal Overload Relay 4.2 Fuses 4.3 Miniature Circuit Breaker (MCB) 4.4 Molded Case Circuit Breaker (MCCB) 4.5 Phase Loss Protection device 4.6 Over/Under Voltage protection relay 4.7 Temperature Sensors
5. Motor	5.1 AC Motor 5.1.1 Induction Motor 5.1.2 Split-phase motor 5.1.3 Capacitor-start motor 5.1.4 Capacitor-run motor 5.1.5 Shaded-pole motor 5.1.6 Three-Phase Induction Motor 5.1.7 Synchronous Motor 5.1.8 Servo Motor 5.2 DC Motor 5.2.1 Brushed DC Motors

	5.2.2 Brushless DC Motor 5.2.3 Coreless DC Motors 5.2.4 Stepper Motors 5.2.5 Servo Motors
6. Starters	6.1 DOL starter 6.2 Star-Delta Starter 6.3 Reverse-Forward Starter

Curricular Content Guide

1. Underpinning Knowledge	1.1. Types of motors (single-phase, three-phase) and their applications. 1.2. Motor control circuits and starter systems (DOL, Star-Delta & Reverse-Forward). 1.3. Electrical symbols, diagrams, and technical drawings related to motor connections. 1.4. Relevant electrical standards and safety regulations. 1.5. Protective devices for motors (e.g., fuses, circuit breakers, thermal overload). 1.6. Principles of motor operation, including voltage, current, and power specifications. 1.7. Grounding and insulation requirements for motor installations. 1.8. Troubleshooting motor connections and common motor faults.
2. Underpinning Skills	2.1 Collecting manuals and diagrams required for motor connection 2.2 Select appropriate controlling and protective devices based on motor specifications. 2.3 Interpret electrical drawings and symbols accurately for motor installation. 2.4 Install controlling and protective devices as per layout plans. 2.5 Connect single-phase and three-phase motors following wiring diagrams. 2.6 Strip insulation and crimp cable lugs securely for motor connections. 2.7 Test motor operation and verify proper functionality after connection. 2.8 Adjust motor rotation direction if needed. 2.9 Apply safety standards and ensure compliance during installation and testing.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness

	3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Materials, tools and equipment needed for the activity 4.3 Instruction manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified and selected controlling and protective devices for motor connection 1.2 installed controlling and protective devices 1.3 installed and operated single phase Induction motor 1.4 installed and operated three phase Induction motor
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: OPERATE THREE PHASE GENERATOR	Nominal Duration: 70 Hrs.	Unit Code: SICIP-SBD-EIM-03-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to operate three phase generators. It specifically includes installing, connecting and repairing automatic transfer switch (ATS), connecting the generator to the main electrical panel, preparing generator for operation, starting and stopping generator, performing post-operation checks and maintenance.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Install, connect and repair Automatic Transfer Switch (ATS)	1.1 ATS is selected based on the power rating and compatibility with the generator and electrical system. 1.2 Schematic diagram of ATS is interpreted 1.3 Connection between the generator and ATS is made according to the manufacturer's guidelines. 1.4 The ATS is tested for smooth transfer between mains power and generator power. 1.5 Faulty ATS is checked and repaired
2. Connect the generator to the main electrical panel	2.1 <u>Components of substation</u> and <u>main electrical panel</u> are described. 2.2 Main electrical panel is inspected to confirm compatibility with the generator. 2.3 Proper cables and connections are selected to handle the load between the generator and substation. 2.4 Connections are securely made with attention to safety protocols and grounding. 2.5 Voltage levels are measured to ensure they match the substation or main panel requirements. 2.6 Testing is conducted to verify smooth power flow from the generator to the substation or panel.
3. Prepare generator for operation	3.1 Generator is placed in a well-ventilated area to avoid fume buildup. 3.2 Fuel level is checked and topped up as required, ensuring the correct fuel type is used. 3.3 Engine oil and coolant levels are inspected and filled to optimal levels. 3.4 Electrical connections, cables, and terminals are checked for wear or damage. 3.5 Specific gravity and level of electrolyte and terminal voltage of Lead acid battery are checked and made ready for use

	3.6 Safety switches and grounding connections are verified before operation.
4. Start and stop generator	4.1 Generator is started following the manufacturer's recommended procedures. 4.2 Warm-up time is allowed as specified for the generator model. 4.3 Load is applied gradually to prevent electrical surges and mechanical stress. 4.4 Generator is monitored for unusual noises, vibrations, or performance issues during operation. 4.5 Proper shutdown procedures are followed, allowing cool-down time before switching off.
5. Perform post-operation checks and maintenance	5.1 Generator is inspected after use to identify any wear, damage, or malfunctions. 5.2 Oil and fuel filters are checked and replaced, if necessary, after extended operation. 5.3 Cables, connectors, and load terminals are inspected and maintained to ensure they remain in good condition. 5.4 Battery (if applicable) is charged or replaced as required for next operation. 5.5 Generator is securely stored, and maintenance logs are updated as per workplace requirements.

Range of Variables

Variable	Range (Includes but not limited to):
1. Components of substation	1.1 Transformer 1.2 High Voltage Switchgear 1.3 Low Voltage Switchgear 1.4 Earthing and Grounding System 1.5 Metering and Protection Systems 1.6 Surge Protection Devices (SPDs)
2. Component of main electrical panel	2.1 Main Incoming Breaker (MCCB/ACB) 2.2 Busbars 2.3 Sub-Circuit Breakers (MCBs/MCCBs) 2.4 Automatic Transfer Switch (ATS) 2.5 Generator Control Circuitry 2.6 Surge Protection Devices (SPDs) 2.7 Energy Meters 2.8 Voltage Stabilizer or Automatic Voltage Regulator (AVR) 2.9 Neutral and Earthing Connection Pit 2.10 Capacitor Banks

Curricular Content Guide

1. Underpinning	1.1 Types and functions of generators and their
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Knowledge	components. 1.2 Operation and function of Automatic Transfer Switch (ATS) and changeover switches. 1.3 Electrical codes and safety regulations for generator installation. 1.4 Power ratings and voltage levels for residential generators. 1.5 Principles of load handling and power distribution. 1.6 Grounding, earthing systems, and electrical protection devices. 1.7 Maintenance procedures for fuel systems, engine oil, and cooling systems. 1.8 Troubleshooting methods for generator malfunctions.
2. Underpinning Skills	2.1 Select appropriate ATS and changeover switches based on power ratings. 2.2 Install ATS and generator connections following manufacturer guidelines. 2.3 Connect the generator to the main electrical panel safely and securely. 2.4 Inspect electrical connections, cables, and terminals for wear or damage. 2.5 Monitor the generator during operation for any unusual behavior. 2.6 Start and stop the generator according to safety protocols. 2.7 Perform post-operation checks and routine maintenance on the generator. 2.8 Test voltage levels to ensure correct power flow
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (actual or simulated) 4.2 Access to relevant work place or appropriately simulated environment where assessment can take place 4.3 Materials and equipment relevant to the proposed activity or task Work place 4.4 Tools and equipment appropriate to joint and connection process. 4.5 Drawings and specifications relevant to the task

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 installed connect and repair automatic transfer switch (ATS) 1.2 connected the generator to the main electrical panel 1.3 prepared generator for operation 1.4 started and stopped generator 1.5 performed post operation checks and maintenance
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM SINGLE PHASE MOTOR REWINDING AND SERVICING	Nominal Duration: 60 Hrs.	Unit Code: SICIP-SBD-EIM-04-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform single phase motor rewinding and servicing. It specifically includes checking single phase motor and detecting the actual fault, selecting tools and preparing material for winding / rewinding, carrying out winding / rewinding of stator, rotor and armature, making connections, carrying out pre-assembly tests and assembling the machine; carrying out final test and recording the test result.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Check single phase motor and detect fault	1.1. Visual and manual inspection is carried out to detect the mechanical damage/defects. 1.2. Polarity, Continuity and short circuit tests are carried out using <u>measuring instrument</u> to detect fault 1.3. Burnt winding is identified. 1.4. The motor is dismantled as per standard procedure 1.5. <u>Winding data</u> is collected and winding diagram is prepared. 1.6. Associated accessories faults are checked and noted down. 1.7. Bearing, carbon brushes, rockers, slip rings are checked visually and tested by specified instrument. 1.8. Fault is detected and noted down for proper repair.
2. Select tools and prepare material for winding / rewinding	2.1 Technical information and winding data is collected and winding diagram is prepared as per winding data 2.2 Specified <u>tools, materials</u> and equipment are selected for winding work. 2.3 Winding wire of required gauge, insulation and binding material are prepared 2.4 Formers are prepared according to winding data. 2.5 Coil is formed on former as per collect winding data.
3. Carryout winding / rewinding of stator rotor and armature	3.1 Stator/rotor and armature are cleaned by using specified tools. 3.2 Insulation material is inserted into slots 3.3 Formed coil is inserted into slots as per standard 3.4 Rotor is cleaned and checked 3.5 Servicing and repaired works is carried out as necessary 3.6 Formed coil is insulated, bonded and inserted into slot

4. Make connections, carry out pre-assembly tests and assemble the machine	4.1 Winding resistance and insulation level is checked by specified test instrument. 4.2 Coil ends are terminated and insulated 4.3 Winding continuity is checked 4.4 Dry running is carried out, performance is checked and remedial action has taken if necessary. 4.5 Insulating varnish is applied to winding and baked it up to recommended temperature in baking oven. 4.6 Assembly of stator, rotor and other parts are assembled.
5. Carry out final test	4.1 Rotor free movement is checked. 4.2 No load / load / locked rotor tests are performed. 4.3 Rotor static and dynamic balancing is carried out.
6. Record the test result	4.5 Test results are documented in the relevant recorded sheet 4.6 Test reports are prepared. 4.7 Machine performance is demonstrated 4.8 Tools, equipment and materials are cleaned and stored as per workplace standard.

Range of Variables

Variable	Range (Includes but not limited to):
1. Measuring Instrument	1.1. Multimeter 1.2. Insulation Resistance Tester (Megger) 1.3. Clamp Meter 1.4. Ohmmeter 1.5. Infrared Thermometer/Thermal Camera 1.6. Tachometer
2. Winding data	2.1 Wire Gauge (AWG or SWG) 2.2 Number of Turns per Coil 2.3 Coil Configuration (e.g., lap, wave) 2.4 Number of Coils 2.5 Number of Poles 2.6 Winding Type (e.g., lap, wave) 2.7 Coil Pitch 2.8 Slot Fill Factor 2.9 Wire Insulation Type 2.10 Voltage Rating 2.11 Current Rating 2.12 Slot Dimensions 2.13 Conductor Material (e.g., copper, aluminum) 2.14 Motor Phase Type (Single-phase, Three-phase) 2.15 Insulation Resistance Value 2.16 Winding Connection
3. Tools	3.1 Flat / Philips headed screw drivers 3.2 Bradawl 3.3 Electrician knife

	3.4 Diagonal cutting pliers 3.5 Combination pliers 3.6 Round nose pliers 3.7 Stripping pliers 3.8 Adjustable wrenches 3.9 Hacksaw 3.10 Hammer 3.11 Files 3.12 Measuring tape 3.13 Steel ruler 3.14 Soldering iron 3.15 Neon line tester 3.16 Spirit level 3.17 Continuity tester. 3.18 Socket wrenches sets 3.19 Bearing puller 3.20 Crimping pliers 3.21 Saw 3.22 Divider 3.23 Micrometer 3.24 Bolt cutter 3.25 Pipe cutter 3.26 Manual hand drill 3.27 Percussion drill set.
4. Materials	4.1 Super enamel wires 4.2 Insulating plastic film 4.3 PVC tape 4.4 Laminates 4.5 Insulating paper 4.6 Ceramic fiber Flexible cables 4.7 Slot insulation paper 4.8 Binding thread 4.9 Varnish 4.10 Thinner 4.11 Grease 4.12 Bearings 4.13 Solder 4.14 Resin 4.15 Winding string

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of single-phase motors: construction, operation, types, and applications 1.2 Estimation of winding. 1.3 Types of winding. 1.4 Process of preparation of insulation paper, coils and wedges.
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	1.5 Setting the insulation paper, coils and wedges. 1.6 Internal position of the coil groups. 1.7 Process of connection of coils (winding) in groups. 1.8 Separation of groups of coils. 1.9 Process of connection of the winding terminals to the terminal post. 1.10 Process of preparation and application of varnish. 1.11 Measurement of insulation resistance
2. Underpinning Skills	2.1 Using the proper tools 2.2 Disassembling the motor 2.3 Tracing out the internal connection of the winding 2.4 Preparation of the insulation paper coils and wedges 2.5 Setting the insulation paper, coils and wedges in the slots 2.6 Testing the connection of the coils with meter 2.7 Assembling the motor for test run 2.8 Preparation of the motor for varnishing 2.9 Assembling the motor for use
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (actual or simulated) 4.2 Access to relevant work place or appropriately simulated environment where assessment can take place 4.3 Materials and equipment relevant to the proposed activity or task Work place 4.4 Tools and equipment appropriate to joint and connection process. 4.5 Drawings and specifications relevant to the task

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 checked single phase motor and detect fault 1.2 selected tools and prepare material for winding / rewinding 1.3 carried out winding / rewinding of stator rotor and armature 1.4 make connections, carry out pre-assembly tests and assemble the machine 1.5 carried out final test 1.6 recorded the test result
2. Methods of	Competency should be assessed by:

Assessment	2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM THREE PHASE MOTOR REWINDING AND SERVICING	Nominal Duration: 70 Hrs.	Unit Code: SICIP-SBD-EIM-05-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform three phase motor rewinding and servicing. It specifically includes checking three phase motor physically and detecting the actual fault, selecting tools and prepare material for winding / rewinding, carrying out winding / rewinding of stator, rotor and armature, making connections, carrying out pre-assembly tests and assembling the machine, carrying out final test and recording the test result.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Check three phase motor and detect fault	1.1. Visual and manual inspection is carried out to detect the mechanical damage/defects. 1.2. Polarity, Continuity and short circuit tests are carried out using <u>measuring instrument</u> to detect fault 1.3. Burnt winding is identified. 1.4. The motor is dismantled as per standard procedure 1.5. <u>Winding data</u> is collected and winding diagram is prepared. 1.6. Associated accessories faults are checked and noted down. 1.7. Bearing, carbon brushes, rockers, slip rings are checked visually and tested by specified instrument. 1.8. Fault is detected and noted down for proper repair.
2. Select tools and prepare material for winding / rewinding	2.1 Technical information and winding data are collected and winding diagram is prepared as per winding data 2.2 Specified <u>tools</u> , <u>materials</u> and equipment are selected for winding work. 2.3 Winding wire of required gauge, insulation and binding material are prepared 2.4 Formers are prepared according to winding data. 2.5 Coil is formed on former as per collect winding data.
3. Carryout winding / rewinding of stator rotor and armature	3.1 Stator/rotor and armature are cleaned by using specified tools. 3.2 Insulation material is inserted into slots 3.3 Formed coil is inserted into slots as per standard 3.4 Rotor is cleaned and checked 3.5 Servicing and repaired works is carried out as necessary 3.6 Formed coil is insulated, bonded and inserted into slot

4. Make connections, carry out pre-assembly tests and assemble the machine	4.1 Winding resistance and insulation level is checked by specified test instrument. 4.2 Coil ends are terminated and insulated 4.3 Winding continuity is checked 4.4 Dry running is carried out, performance is checked and remedial action has taken if necessary. 4.5 Insulating varnish is applied to winding and baked it up to recommended temperature in baking oven. 4.6 Assembly of stator, rotor and other parts are assembled.
5. Carry out final test	5.1 Rotor free movement is checked. 5.2 No load / load / locked rotor tests are performed. 5.3 Rotor static and dynamic balancing is carried out.
6. Record the test result	6.1 Test results are documented in the relevant recorded sheet 6.2 Test reports are prepared. 6.3 Machine performance is demonstrated 6.4 Tools, equipment and materials are cleaned and stored as per workplace standard.

Range of Variables

Variable	Range (Includes but not limited to):
1. Measuring Instrument	1.1. Multimeter 1.2. Insulation Resistance Tester (Megger) 1.3. Clamp Meter 1.4. Ohmmeter 1.5. Infrared Thermometer/Thermal Camera 1.6. Tachometer
2. Winding data	1.1 Wire Gauge (AWG or SWG) 1.2 Number of Turns per Coil 1.3 Coil Configuration (e.g., lap, wave) 1.4 Number of Coils 1.5 Number of Poles 1.6 Winding Type (e.g., lap, wave) 1.7 Coil Pitch 1.8 Slot Fill Factor 1.9 Wire Insulation Type 1.10 Voltage Rating 1.11 Current Rating 1.12 Slot Dimensions 1.13 Conductor Material (e.g., copper, aluminum) 1.14 Motor Phase Type (Single-phase, Three-phase) 1.15 Insulation Resistance Value 1.16 Winding Connection
3. Tools	3.1 Flat / Philips headed screw drivers 3.2 Bradawl

	3.3 Electrician knife 3.4 Diagonal cutting pliers 3.5 Combination pliers 3.6 Round nose pliers 3.7 Stripping pliers 3.8 Adjustable wrenches 3.9 Hacksaw 3.10 Hammer 3.11 Files 3.12 Measuring tape 3.13 Steel ruler 3.14 Soldering iron 3.15 Neon line tester 3.16 Spirit level 3.17 Continuity tester. 3.18 Socket wrenches set 3.19 Bearing puller 3.20 Crimping pliers 3.21 Saw 3.22 Divider 3.23 Micrometer 3.24 Bolt cutter 3.25 Pipe cutter 3.26 Manual hand drill 3.27 Percussion drill set.
4. Materials	4.1 Super enamel wires 4.2 Insulating plastic film 4.3 PVC tape 4.4 Laminates 4.5 Insulating paper 4.6 Ceramic fiber Flexible cables 4.7 Slot insulation paper 4.8 Binding thread 4.9 Varnish 4.10 Thinner 4.11 Grease 4.12 Bearings 4.13 Solder 4.14 Resin 4.15 Winding string

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of three-phase motors: construction, operation, and applications. 1.2 Types of three-phase motors: squirrel cage, wound rotor, and synchronous motors. Estimation of winding. 1.3 Types of winding.
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	1.4 Process of preparation of insulation paper, coils and wedges. 1.5 Setting the insulation paper, coils and wedges. 1.6 Internal position of the coil groups. 1.7 Process of connection of coils (winding) in groups. 1.8 Separation of groups of coils. 1.9 Process of connection of the winding terminals to the terminal post. 1.10 Process of preparation and application of varnish. 1.11 Measurement of insulation resistance
2. Underpinning Skills	2.1 Diagnosing faults in three-phase motors using appropriate tools and test instruments. 2.2 Using the proper tools 2.3 Disassembling the motor 2.4 Tracing out the internal connection of the winding 2.5 Preparation of the insulation paper coils and wedges 2.6 Setting the insulation paper, coils and wedges in the slots 2.7 Testing the connection of the coils with meter 2.8 Assembling the motor for test run 2.9 Preparation of the motor for varnishing 2.10 Assembling the motor for use
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (actual or simulated) 4.2 Access to relevant work place or appropriately simulated environment where assessment can take place 4.3 Materials and equipment relevant to the proposed activity or task Work place 4.4 Tools and equipment appropriate to joint and connection process. 4.5 Drawings and specifications relevant to the task

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 checked three phase motor and defect fault 1.2 selected tools and prepared material for winding/re-winding 1.3 carried out winding/re-winding of stator rotor and armature 1.4 make connections carried out pre-assembly tests and
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	assemble the machine 1.5 carried out final test 1.6 recorded the test result
2. Methods of Assessment	Competency should be assessed by: 2.1 Written examination 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio review
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

End of the Competency Standard

Workshop/Lab Facility Standard

Course Name:	Electrical Installation and Maintenance (Shipbuilding)
Number of Trainees:	25

Course-wise Training Space (Theoretical Classroom, Workshop/ Lab/ Classroom cum Workshop):

- Classroom – 350 sft (33 sqm)
- Workshop/ lab – 800 sft (75 sq) OR
- Classroom cum workshop – 1000 sft (93 sqm)

Major Training Equipment and Training Facilities:

Sl. No.	Major Equipment and Training facilities	Required facilities
1.	Wiring Boards/Trainer Board	12
2.	Navigation lights	5
3.	Echo sounder	5
4.	Magnetic compass	5
5.	GPS recorder	
6.	Main Distribution Board with busbar, CB and other accessories (Training model)	5
7.	Sub Distribution Board with busbar, CB and other accessories (Training model)	5
8.	Electric drill machine	5
9.	Electric grinding machine	5
10.	Energy Meter (1 Phase)	5
11.	Energy Meter (3 Phase)	5
12.	Earth tester	2
13.	Megger	2
14.	Phase sequence meter	1
15.	Three phase motor (5-10HP)	8
16.	Single Phase Motor	5
17.	Burnt/Defective motor	4
18.	ATS	5
19.	DOL Starter	5
20.	Star-Delta Starter with controlling arrangement	5
21.	Generator (15-25KW)	1
22.	Lead Acid Battery (12 V, 100 AH)	2
23.	Forward-Reverse circuit arrangement for motor control	5
24.	Wiring Boards/Trainer Board	12

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on Electrical Installation and Maintenance the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.